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Prediction and quantification of bacterial biofilm detachment using Glazier–Graner–Hogeweg method based model simulations

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ABSTRACT

Morphological changes in bacterial biofilm structures arise from the fluid-structure interactions between the biofilm and the surrounding fluid. Depending on the magnitude of the force acting on the structure, the bacteria rearrange to attain an equilibrium shape or get washed away by the moving fluid. Understanding the dynamics behind the evolution of such equilibrium or failed states can aid in development of tools for biofilm removal or eradication. We develop a Glazier–Graner–Hogeweg method-based model to explore the collective evolution of biofilm morphology arising from cell-cell and cell-fluid interactions. We show that low adherence and high motility of the cells leads to sloughing of biofilms. Also, streamers are found to form under laminar flow conditions in tightly packed biofilms. In mixed species biofilms, we found that a species with less cell-cell binding affinity gets eroded faster than its counterpart. Therefore, we hypothesize that in nature these less-adherent species should be present encapsulated within the biofilm structure to maximize their chances of survival.

1. Introduction

Bacterial biofilms consist of structures that house millions of bacteria that stay as a single colony. The bacteria in the biofilms are different from planktonic cells, the free roaming bacteria, in that they are confined within the biofilm and are seldom motile. The community members (bacteria) interact at microscopic levels with each other and the ambient environment to define the overall evolution of the system (Sutherland, 2001). The biofilms offer cover to the confined bacteria from antibiotics (Roberts and Stewart, 2005; Mah and O'toole, 2001), bacteriophages, and external stresses. Due to their strong adherence to the substratum, they are hard to remove or eradicate, thus leading to biofouling and microbial induced corrosion (MIC) of the adhering surface. Biofouling of pipes used in heat exchangers (Liu et al., 2017) or offshore structures (Vigneron et al., 2018) may lead to pressure drop, leading to system failure and damage. Bacterial biofilms are also a major concern in Fast Breeder Test Reactor (George et al., 2016) or other nuclear reactors (Giacobone et al., 2015) due to their presence in the cooling water systems. Any uncontrolled build-up of biofilms within these systems will result in devastating consequences. In cases of water treatment plants, growth of biofilms decreases the flux of treated water, necessitating increase of pressure within the system (Fortunato et al., 2017; Vrouwenvelder et al., 2009) thus adding to operational costs. To prevent such outcomes, bacterial biofilms must be periodically removed from the systems. The mechanical process of biofilm removal is known as detachment, where a part or whole of the biofilm is removed from its original site due to mechanical stresses from flowing fluid and localized structural defects occurring within the biofilm. In most industries, detachment of biofilms is carried through high pressure washing or mechanical scrubbing (Jessen and Lammert, 2003). The efficiency of such methods depends on parameters such as the type of the biofilm, the location of biofilm within the structure and device geometry.

In literature, there have been simulation studies to estimate the detachment rate of biofilms in various systems. Quantifying the detachment process started with modelling the rate of detachment as a function of biofilm thickness (Stewart, 1993; Peyton and Characklis, 1993) or biomass density, ignoring the spatial effects of the

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biofilm structure. There have been models that address the spatial detachment process using stochastic variables dependent on the height of the biofilm or relative position of the bacteria in a biofilm (Li et al., 2015). Picioreanu et al. (2001) developed a more realistic shear stress-based detachment model assuming the biofilm as an elastic structure. They simulated 2D flow in a channel comprising the biofilm and evaluated the detachment rate of the bacteria with time. Following a similar approach, Tierra et al. (2015) developed a three- phase multicomponent model, where they considered biofilm as an incompressible viscous fluid that deforms from interaction with the ambient forces. However, treating the entire biofilm as a single mechanical structure with homogenous physical and biological properties does not truly represent the heterogenous nature biofilm's mechanical properties. It has been shown that the physical properties such as tensile strength vary throughout the biofilm based on the localized bacterial populace and extracellular polymeric substances (EPS) distribution. To capture such cell level dependencies of the mechanical properties of biofilm, Jayathilake et al. (2017) implemented mechanistic interaction between the cells using discrete element method. They assumed a fixed shear rate profile for the fluid interacting with the biofilm, which composed of hard spherical particles. This fluid-particle interaction in the model was used to predict the roughness and detachment rates of the biofilm formed. All these models, however, do not account for the localized interactions arising from the cell motility and cell shape fluctuations, which are important determinants of the biofilm's response (Rupp et al., 2005) to the eradication techniques. These cell level activities can give rise to formation of mechanically interesting structures such as streamers (Stoodley et al., 1999), piece-wise detachment (Stoodley et al., 2002) of the structure and further colonization of the neighbourhood (Rupp et al., 2005). Therefore, bacterial biofilms are community-driven heterogenous biological systems, which warrant the use of Agent-Based simulations to explore their emergent properties (Flemming et al., 2016). Biological cell-cell interactions have been modelled using agent-based models (Railsback and Grimm, 2011; Gilbert, 2008) since they can handle individual attributes of the cells and simulate the evolution of a larger system which encompasses these cells. Frameworks such as NetLogo (Wilensky and Rand, 2015), FLAME (Holcombe et al., 2006; Li et al., 2013) and MASON (Luke et al., 2005) offer a wide range of tools to develop agent-based models. These frameworks can handle a wide range of problems in economics, socio-political and geographical domains, which require and agent-based approach. However, to handle individual biological cells in the model, along with their associated physical characteristics such as cell shape, intercellular adhesion and cell signaling, a robust biophysics-based model is required. iDYnomics (Lardon et al., 2011), an individual Based Model framework (iBM), is one such biophysics-based model, which simulates cells as hard spheres with an adhesion potential serving as a mechanical link between these cells. The adhesion potential varies as a function of the distance between the cells. iDYnomics was developed with primary focus on simulating bacterial communities and their associated surroundings. Jayathilake et al. (2017) developed a flexible iBM model based on Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) to simulate the adhesion between the cells using springs and dashpot model. The major drawback of using the above iBM based models for simulating bacterial communities is the absence of proper mechanisms to incorporate the dynamics associated with cell shape variations, topographical cell stacking (Sheraton et al., 2018), and the consequent effects associated with cell-cell interactions. Glazier–Graner–Hogeweg model (GGH) (Popławski et al., 2008) eliminates these shortcomings by implementing energy-based evolution of cell shapes (Akanuma et al., 2016; Li and Lowengrub, 2014). In addition, the energy-based approach restricts the evolution of physically unrealistic scenarios which would otherwise arise in other simplistic approaches (Jayathilake et al., 2017; Lardon et al., 2011), which assume cells to be a fixed solid geometry such a sphere. The proliferation of cells in GGH is also indirectly governed by the energy minimization principle, thus preventing artificial growth dictated by the nutrient diffusion-reaction equations. GGH model has been used in simulating various cell level and tissue level phenomena such as avascular tumor growth, angiogenesis (Shirinifard et al., 2009), and biofilm simulations (Sheraton et al., 2018; Popławski et al., 2008). Due to its ability to handle multiple cell types, each with their own interacting energies, GGH model can be readily used to simulate multispecies bacterial communities and somitogenesis (Hester et al., 2011).

GGH model-based simulations can include all cell level activities and incorporate external force coupling with the cell through the Hamiltonian and Non-Hamiltonian energy (Glazier et al., 2007) potential associations. In this paper, we develop a hybrid GGH – Finite Element Method (Versteeg and Malalasekera, 2007) based model to understand the effect of shear stress on the shape of the biofilm and to quantify the loss of biomass due to detachment. We analyze the potential outcomes of the detachment process, based on the strength of the biofilm or the adhesion strength between individual bacteria. Since there are numerous models in literature (Wang and Zhang, 2010) detailing the bacterial cell proliferation and nutrient based growth of biofilms, we initialize the model with a fixed structure such the classical mushroom shape, ignoring the evolution of biofilm before the initiation of the removal processes. We then examine the influence of various biophysical model parameters such as, shear stress, adhesion potential and motility on the morphology of the biofilm structure.

2. Model description

2.1. Model for bacterial cells

We use GGH model (Glazier et al., 2007; Marée and Hogeweg, 2001; Balter et al., 2007; Graner and Glazier, 1992; Glazier and Graner, 1993; Belmonte et al., 2016) to model the bacterial cells in the biofilm. To implement the GGH model, we use the opensource "CompuCell3D 3.7.5" software framework (Popławski et al., 2008; Swat et al., 2012). The simulations are implemented in a 2-Dimensional GGH domain. The individual bacteria are modelled as a collection of pixels. The spatial location of a pixel is denoted by $\vec{i}$ and the associated object (cell) is denoted by σ. Each pixel has a fixed length 'dx' in space. The volume of the cells 'V_{cell}' is set equal to the average volume of *Pseudomonas aeruginosa* cells (Fagerlind et al., 2012). The volume of the cells in the simulations is governed by the volume constraint equation (Eq. (1)). In the equation, E_V is the energy associated with volume, λ denotes the volume potential, V_T denotes the target volume and V_{cell} is the volume of the cell.

$$E_V = \sum_{\sigma} \lambda (V_{cell}(\sigma) - V_T(\sigma))^2 \quad (1)$$

The adhesion between different cells in the simulation is dictated by the contact energy (E_c) constraint shown in Eq. (2).

$$E_c = \sum_{i,j} J\left(\tau_{\sigma(i)}, \tau_{\sigma(j)}\right)\left(1 - \delta_{\sigma(i),\sigma(j)}\right) \quad (2)$$

where, τ indicates the type of the cell at the location '$\sigma(i)$' or '$\sigma(j)$, J denotes the contact energy potential between two cell types, δ is the Kronecker delta function, which equals 1 if $\sigma(i)$ and $\sigma(j)$ are equal, else the value equals 0. In this study, the value of contact energy potential ($J(\tau_{\sigma(i)}, \tau_{\sigma(j)})$) between two different cell types X and Y is denoted by C_{XY}. In addition to the bacterial cells, we model the substratum as immobile (frozen) group of cells with fixed volume (Swat et al., 2013). These cells form the bottom layer on which the lowest layer of bacterial cells in the biofilm rest and adhere to. The adhesion between bacteria and substratum is made stronger than the bacteria-bacteria adhesion (C_{BB}), which is reflected in the model by a lower adhesion potential (C_{BS}) parameter for the bacteria-substratum interactions. The substratum layer introduces frozen boundary conditions (Swat et al., 2013) at the bottom of the simulation domain. No-flux boundary conditions are used along the other edges (top, left and right) of the GGH domain. In addition to the volume and contact constraints, a connectivity constraint is included in the model. The connectivity constraint E_p acts as a penalty parameter which prevents the cells from fragmentation, due to numerical instabilities (Swat et al., 2012).

The normal and shear forces experienced by the cells are modelled as a field around the cells. Since the volume of individual pixels is very small compared to the size of the domain and the cells themselves, the forces are applied over the surface of the pixels. If a part of a cell is exposed to the flowing fluid, application of forces to individual pixels ensures that the forces are applied to that localized region instead of the entire cell surface. Eq. (3) describes the energy arising from the displacement of cell pixels ($|\overrightarrow{i}' - \overrightarrow{i}|$) by the GGH pseudo-force ($\overrightarrow{F_{GGH}}$ or GGH F). The GGH pseudo-force in the model is directly proportional to the stresses per unit area of pixel (Traction, $\vec{T}$), developed by the biofilm structure in response to the ambient fluid flow. (The overall implementation of the module in Compucell3D is done similar to the 'ExternalPotential' plugin (Swat et al., 2013)) The traction force $\vec{T}$ is converted to equivalent force $\vec{F}$, which acts along the entire area (S_A) of the pixel. This equivalent force is converted to pseudo GGH force, GGH F, using the proportionality constant k_{GGH}, as shown in Eq. (4b)

$$E_f = \sum_i \overrightarrow{F_{GGH}}\ (i).\left(\left|\vec{\iota}' - \vec{\iota}\right|\right) \tag{3}$$

$$\overrightarrow{F_{GGH}} \ \propto \ \vec{T} \tag{4}$$

$$\vec{T} = \frac{\vec{F}}{S_A} \tag{4a}$$

$$\overrightarrow{F_{GGH}} = k_{GGH}\ \vec{F} \tag{4b}$$

The cells have inherent motility based on the membrane fluctuations, modelled by the temperature term 'T_m' in GGH model. At each simulation time step, one of the pixels (target pixel) in the domain is chosen at random and the state (cell) of a randomly picked (source pixel) lattice site in its neighborhood is copied, this process is known as pixel copy attempt. If the target and destination lattice sites are of the same cell, then there is no pixel copy. A Hamiltonian is calculated at every copy attempt, to determine the energy of the cells and the system. The Hamiltonian H includes the energies originating from volume constraints, adhesion potentials, connectivity penalty and pseudo-force potentials. The probability of success for a pixel copy attempt or in this case, the bacterial membrane fluctuation, is dictated by Eq. (5).

$$P\left(\sigma\left(\vec{\iota}\right) \rightarrow \sigma\left(\vec{\iota}'\right)\right) = \begin{cases} \left[\exp\left(-\frac{\Delta H}{T_m}\right)\right], \Delta H > 0 \\ 1, \Delta H \leq 0 \end{cases} \tag{5}$$

The term, ΔH is the change in energy due to the pixel copy event for the cells at positions $\sigma(\vec{i})$ to $\sigma(\overrightarrow{i'})$. The larger the value of T_m, the faster the membrane fluctuations occur.

2.2. Model for stress calculations

To estimate the traction force ($\vec{T}$) and consequently the normal (σ_n) and shear stresses (τ) arising from fluid interaction with the biofilm surface, we solve the continuity equation and incompressible Navier-Stokes (NS) equation in a 2D channel of dimensions x' × y'. An opensource finite element method-based solver, FENICS, is used for solving the continuity and NS equations. Incremental Pressure Correction Scheme (IPCS) (Valen-Sendstad et al., 2012) is used for the simulation of the transient state fluid flow. The biofilm is considered as a solid object along the flow path and therefore no-slip boundary conditions are applied along the surface the biofilm. The inlet velocity is fixed as a parabolic velocity profile at the channel inlet (at $x = 0$) as shown in Eq. (6), where $\vec{u}$, $\vec{v}$ and $\overrightarrow{u_{max}}$ are velocity in x-direction, velocity in y-direction and maximum inlet velocity respectively. a and b are constants similar to the study by Tierra et al. (2015).

$$\vec{u} = \frac{\overrightarrow{u_{max}}}{a}\left(bY - Y^2\right), \vec{v} = 0, Y = \frac{y}{b}; \tag{6}$$

$$p = 0; \tag{7}$$

Atmospheric pressure is fixed at the exit boundary (at x = x', Eq. (7)), the equations are solved transiently at discrete time intervals, $\Delta t = 4$ min. A very low Reynold's number, $Re \ll 1$, similar to other experimental biofilm studies (Rusconi et al., 2011; Valiei et al., 2012; Drescher et al., 2013), is maintained in all model simulations. From the velocity and pressure values the traction force $\vec{T}$ acting on the surface of the biofilm is calculated from the stress tensor σ using Eq. (8).

$$\vec{T} = \sigma.\vec{n} \tag{8}$$

The traction force is split into individual stress components acting on the surface, namely normal stress σ_n and shear (or tangent) stress τ. Normal stress acts along the normal direction $\vec{n}$ to the surface of the biofilm and shear stress acts parallel to the surface $\vec{s}$. The calculated stresses are applied as an equivalent force (Eq. (4a)) $\vec{F}$, which is then translated to GGH pseudo force ($\overrightarrow{F_{GGH}}$ *or GGH F*) on the individual pixels of the cells, whose surface is exposed to the incoming fluid.

$$\sigma_n = \vec{T}.\vec{n} \tag{9}$$

$$\tau = \vec{T}.\vec{s} \tag{10}$$

The GGH pseudo-force is applied along the surface of the biofilm for approximately every 600 mcs depending on the numerical stability of the flow simulations and meshing. The mcs can be related to the real-time, based on the discrete time unit, Δt, used in the fluid flow simulations. We assume that the changes in shape of the biofilm due to the application of force does not alter the flow patterns and the shear stresses significantly within this interval (600 mcs = 4 min or 1 mcs = 0.4 s). Therefore, the steady state stresses are calculated at time 't' and the resultant GGH forces are applied on the cells from t → t+dt in the GGH interval (dt ~ 600 mcs). During this interval, the bacteria in the model respond to forces acting on the biofilms surface based on the energy changes ΔH governed by Eq. (5). After t+dt, the fluid flow simulations are run with the new shape configuration and the cycle is repeated until t_{max}. The entire simulation accounts for about 60 min in real time or 9000 mcs. Bacterial proliferation during this time period of 60 min should be minimal. Therefore, we neglect any bacterial proliferation in the simulations and assume that bacterial proliferation does not significantly change the fluid flow patterns throughout the simulations.

2.3. Generation of meshes for fluid dynamics

We generate the meshes for different biofilm shapes at different time instances using GMSH. The generation of meshes is an automated process using the lattice site data of individual cells from CompuCell3D output. We use adaptive mesh refining to make the

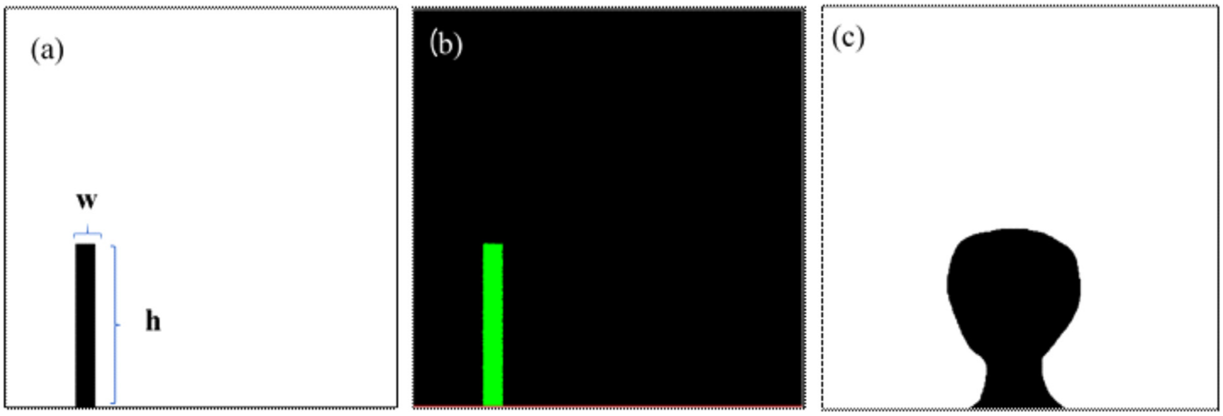

Fig. 1. Generation of biofilm lattice and mesh for the numerical simulations. (a) Portable Network Graphics file (PNG) describing the shape of the structure, (b) Compucell3D cell lattice recreated from the PNG file and (C) Portable Network Graphics file describing the shape of mushroom-shaped biofilm structure. Here, h is the height of biofilm, which is 405 lattice sites and w is the width of biofilm, which is 100 lattice sites.

mesh finer near biofilm surfaces and coarser elsewhere as shown in Fig. 1c. An arbitrary stump-like biofilm shape configuration (Fig. 1a) similar to the one used by Tierra et al. (2015) is drawn as an outline Portable Network Graphics (PNG) file. Mushroom-shaped biofilm (Fig. 1c), a typical structure of biofilms found in the wild (Sheraton et al., 2018; Klausen et al., 2003; Yang et al., 2009), is also used as the starting configuration for the simulation. The outline image is initialized in CompuCell3D and discretized to individual bacterial cells occupying the marked biofilm area. The cells initialized from the image are square shaped and a few may not have their target volume V_T at the time of initialization. Therefore, the simulation is let to anneal for the first 600 mcs, during which the local energy of the cells is minimized, and they attain a uniform volume and shape closer to their target volume V_T by the end of annealing steps.

3. Results and discussion

After the annealing process, the cells in the simulation reach their target volume and the external force is applied on these cells. As the simulation proceeds, the biofilm structure starts to deform based on the stresses acting on its surface. Thus, the forces applied along the periphery are transformed into an energy perturbation traversing through the biofilm volume. The model in turn adjusts the position of the cells within the biofilm to minimize the local energy and consequently the final structure of the biofilm evolves with time. However, since the applied force is a pseudo-force, there are multiple outcomes possible with the change in values of the proportionality constant k_{GGH}. Physically, $k_{GGH}\vec{F}$ or $\overrightarrow{F_{GGH}}$, represents the strength of the force acting on the biofilm surface for 1 mcs and the order of magnitude 'O' of this force impacts the shape of the biofilm structure. In addition to the applied forces, the change in morphology of biofilm structure can be affected by two other model parameters namely, the membrane fluctuations T_m and the contact energy between the bacterial cells [C_{BB}]. These three parameters were therefore varied to understand their effects on the biofilm structure. The results from the parametric variation study are summarized in Fig. 2. The volume changes for each case of parameter variation (Fig. A1) clearly shows that there are no numerical instabilities in the GGH model and that the cells are not swelled up or crushed due to the force applied on them. Three distinct morphological outcomes are observed in Fig. 2, (i) bending, (ii) distortion and (iii) detachment. The process of distortion occurs when the biofilm changes its shape without much loss of its individual members. This loss of bacterial cells from biofilm occurs due to their displacement by fluid shear. Heavy distortion of structures is commonly observed in biofilms with high contact energy as visible in Fig. 2c and d. A high contact energy between the cells translates to less adhesion between the cells. The surface roughness of these distorted structures is very high compared to other biofilms in the simulation. This could mean that biofilms that do not secrete large quantity of EPS during their lifetime or biofilms that have loosely packed structures are bound to be heavily deformed by the incoming flow. In general, with the progress of simulation, majority of the deformed structures get removed from their site due to fluid stress. This process is defined as detachment. From the graphs we observe two types of detachments originating in the simulations. They are removal of chunks of biofilm knows as sloughing and wash-away of individual cells in small quantities known as erosion. Erosion is a common characteristic observed in all the biofilm simulations. Erosion starts only after a considerable time since the introduction of fluid stresses. This is due to the structure getting progressively weakened by fluid flow. If erosion were to be imminent once the fluid is introduced in simulation, then the cells would be outright plucked from the biofilm by fluid force, but this is not the case. As expected, the structure experiencing high GGH forces (Fig. 2a) is more readily weakened, with erosion starting as early as 2000 mcs. But surprisingly, the structures are also weakened swiftly when their members possess high membrane fluctuation values (T_m). High values of membrane fluctuation translate to faster motility of the cell. Increased motility should have made it easy for the biofilm to adjust its structure swiftly in response to the forces experienced by its cells, instead it has resulted in an unstable configuration. This could be a side effect of the model's basic formulation which combines motility with high energy probabilities (Eq. (1)). Minimization of the energy of the system could result in minimized external force experienced by the cells. Therefore, increase in the fluctuation amplitude will result in increased frequency of non-optimal positioning of the cells against the external field. This results in increased erosion of cells which are directly exposed to the incoming fluid.

Sloughing follows a similar pathway as erosion, removal of cells due to external force, except it is characterized by critical or total capitulation of the structural integrity as seen in Fig. 2a, c, d, f and g. There are simulations in which sloughing has occurred in a biofilm more than once such as in Fig. 2a. This shows that there are multiple tipping points in the biofilm's structural integrity, which occur based on the local internal stress effects originating

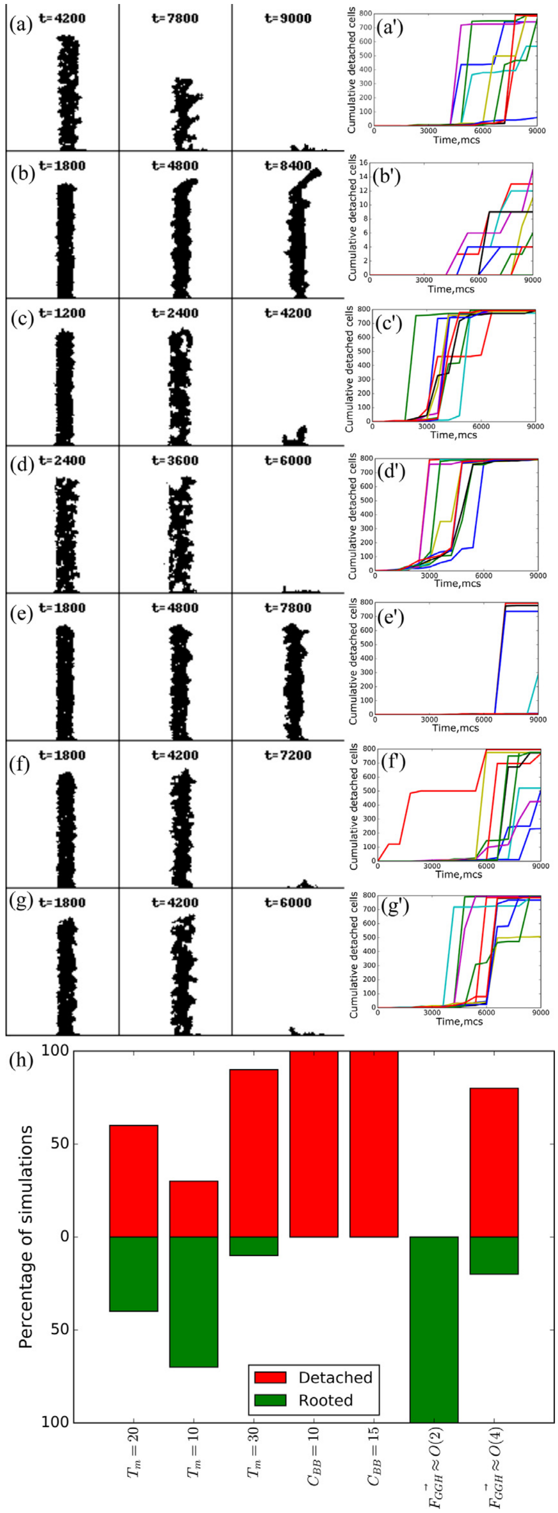

Fig. 2. Effect of fluid stresses acting on the biofilm structure. The snapshots show the structural deformation at different time instances and the xy plots are detachment plots that show the cumulative detached cell count with time for simulations with $T_m = 20$, C_{BB} =5 and (a, a') $\overrightarrow{F_{GGH}} \approx O(4)$, (b, b') $\overrightarrow{F_{GGH}} \approx O(2)$; $T_m = 20$, $\overrightarrow{F_{GGH}} \approx O(3)$ and (c, c') C_{BB} =10, (d, d') C_{BB} =15; $\overrightarrow{F_{GGH}} \approx O(3)$, C_{BB} =5 and (e, e') $T_m = 10$, (f, f') $T_m = 20$ and (g, g') $T_m = 30$. Plot (h) shows the percentage of detached and rooted (attached to substrate) biofilms structures for different parametric values of T_m, C_{BB} and $\overrightarrow{F_{GGH}}$. (The values $T_m = 20$, C_{BB} =5 and $\overrightarrow{F_{GGH}} \approx O(3)$ are used as base sets and one of these values is changed in simulations. Ten replicates were carried out for each of the parameter settings.).

within the biofilm. Sloughing effects are incredibly strong in biofilms with low cell-cell adhesivity (Fig. 2c and d) due to the inherent structural instability arising from loose packing of cells. In loosely packed structures, almost entire biofilms have been detached in very early stages, as early as <3500 mcs. Sloughing has been absent in majority of simulations with low GGH force and low membrane fluctuations as shown in Fig. 2h.

To gain a better understanding of the sloughing phenomena, the velocity vectors and pressure contours are shown for a biofilm at the verge of sloughing in Fig. 3. For this case, the simulation parameters are $T_m = 20$, C_{BB} =5 and $\overrightarrow{F_{GGH}} \approx O(4)$. From plots 3a and b, it is clear that the velocity of the fluid near the head of the biofilm is relatively much higher than elsewhere in the biofilm. This could mean a higher force acting on the top surface compared to the bottom, which leads development of high localized stresses in this area. The only way to minimize this localized high stress is for the cells to move down or climb up the structure. Since the space above the zenith of the structure is devoid of cells and is low pressure, the cells will try to climb up, rather than moving down. But moving up the biofilm would mean fewer cells at the bottom and reduced stability of the structure. Therefore, the structure will start to detach at weak spots, where the cell density is at its lowest. Thus, there are multiple failure points in the biofilm structure, from which sloughing can initiate and in general, these points correspond to locations with low cell density.

In contrast to completely detached biofilms, in couple of cases, we find that majority of the biofilm structure has been left intact by the fluid flow (Fig. 2b and e). These cases exhibit a bending structure, with curved a bottom and top, which is similar to the behaviour of viscoelastic fluid under low external shear. The biofilms in these cases are elongated than their counterparts due to appearance of slender elongated structures at their top. These protrusions of single or multiple cells forming at the top of these biofilms are similar to streamers that appear in bacterial biofilms growing in an environment with continuous flow. The shape of the streamers allows them to be hydro-dynamically stable in the flowing fluid by reducing the external forces experienced by the cells within the streamer structure. Fig. 4d, e and f capture the formation process of a streamer from the biofilm structure. As seen in Fig. 3b and c, the velocity magnitude is higher around the zenith of the biofilm (near the streamer), which could further bend the streamer and elongate it. Thus, the structure remains stable since the cells move horizontally along the streamer, parallel to the flow direction. This cell migration pattern is in stark contrast to the sloughing case where the cells could only ascend due to the absence of horizontal elongated protrusions. These observations suggest that the best way to eradicate a streamer is through back washing, whereby the pressure is reversed. A reversal in flow would snap the streamer at its base, where the cell density is very low.

All the discussions with regards to the arbitrary stump-like structure simulations was centred around a homogenous bacterial biofilm population. In natural environments, biofilms seldom exist as homogenous entities made of same species of bacteria. They exist as multispecies communities with each specie possessing its own biological and mechanical characteristic. In order to capture the response of such heterogenous biofilm systems to fluid shear, we carried out simulations on a mixed species biofilm modelled akin to the classical biofilm mushroom-shaped structure shown in Fig. 5c. The same base parameters ($T_m = 20$, C_{BB} = 5 and $\overrightarrow{F_{GGH}} \approx O(3)$) used in the stump-like structure simulations were used for the new simulations. Three different cases were studied using the mushroom-shaped structure simulations, (i) a homogenous population, (ii) less adherent and (iii) more adherent cap mixed species populations. The less adherent species used in the simulation are

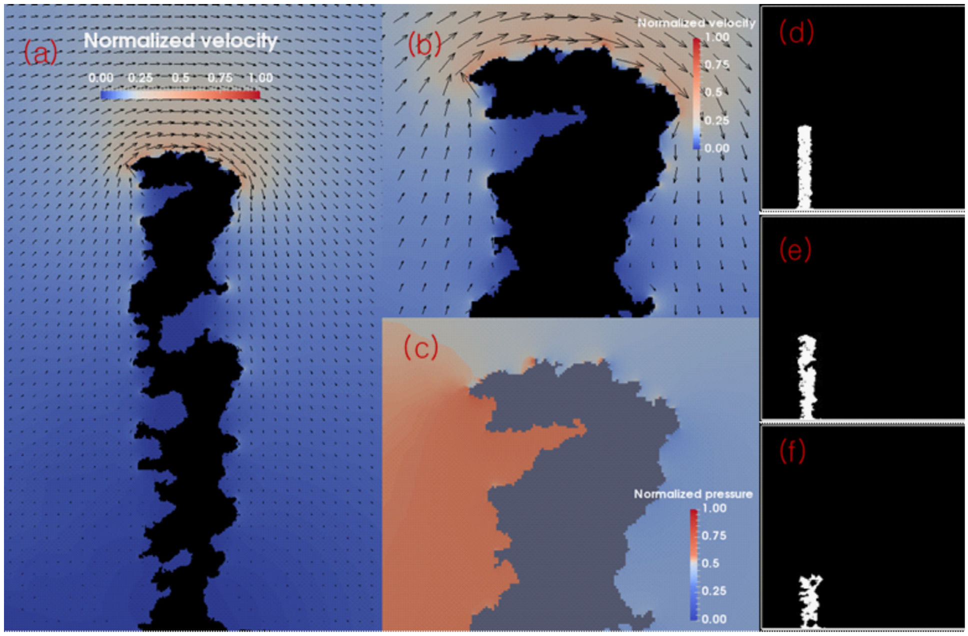

Fig. 3. Fluid dynamic parameters around a biofilm about to be detached. Normalized velocity vector plot for (a) entire biofilm and (b) at the top of the biofilm. (c) Normalized pressure values at the top of the biofilm. Progress of biofilm detachment at (d)1200 mcs, (e) 4200 mcs and (f) 7200 mcs.

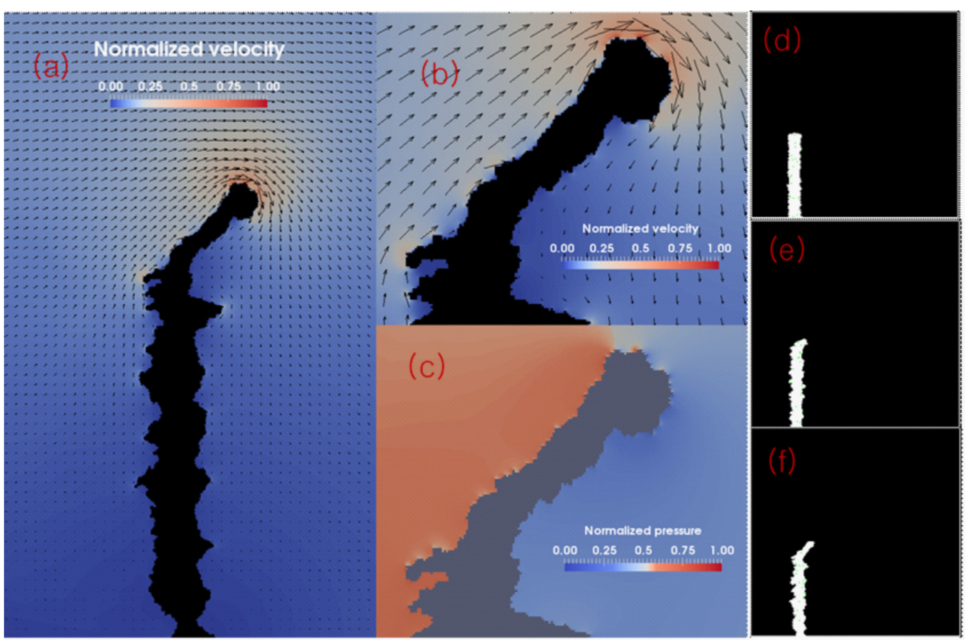

Fig. 4. Fluid dynamic parameters around a biofilm with streamers. Normalized velocity vector plot for (a) entire biofilm and (b) near the streamers. (c) Normalized pressure values near the streamers. Progress of biofilm streamer formation at (d)1200 mcs, (e) 4200 mcs and (f) 7200 mcs.

denoted as B* and more adherent species is denoted by B. The contact energy values between these species are listed in Table 1.

The results from the multispecies biofilm simulations are summarized in Fig. 5. In general, all the mushroom structures, homogenous and heterogenous, suffered structural deformation due to the fluid flow to varying degrees. The bottom parts or the stalks of the mushrooms, which face the incoming fluid, were displaced from their original position in all the cases. The effect of fluid force is more pronounced at the cap of the mushroom, where the structure is heavily distorted. On the leeward side, the peripheral cells are found to move faster than any other cells in the structure, further confirming the propagation of stress effects through the structure. This increased motility is observed in the motility plots of Fig. 5a^{+}, b^{+} and c^{+}. Erosion is predominantly seen to occur in species B* due to their low binding affinity with each other and members of species B. It is important to note that when species B* occupies the bottom as in Fig. 5b, the stalk thins out at a rapid pace compared to the structure with species B occupying the stalk (Fig. 5c). This means that structural configuration in Fig. 5b is inherently unstable and can be completely sloughed away earlier than its

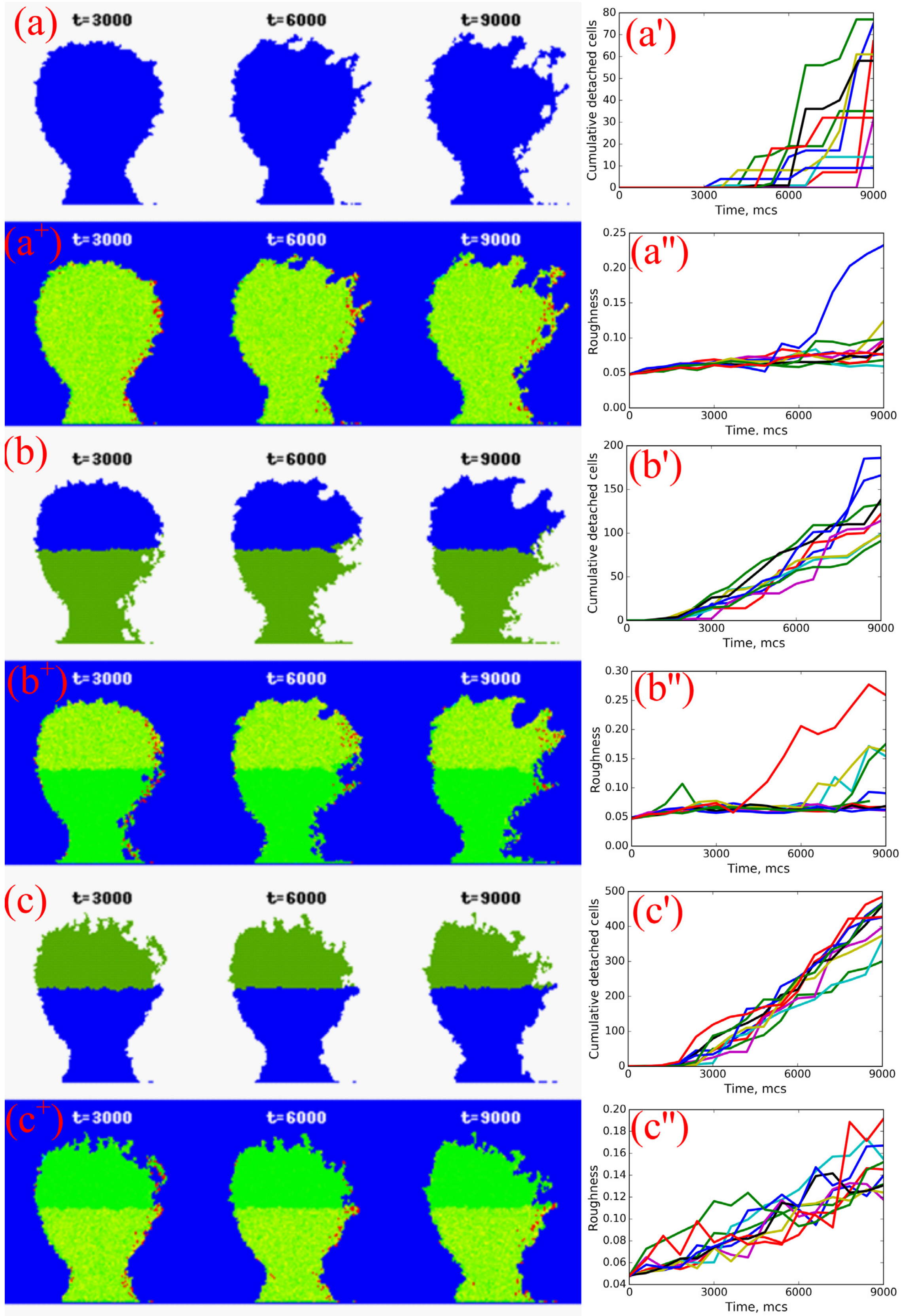

Fig. 5. Effect of fluid stresses acting on the mushroom-shaped biofilm structure. The snapshots show the structural deformation at different time instances ($T_m = 20$, C_{BB} =5, $C_{B^*B^*}$ =15 and $\overrightarrow{F_{GGH}} \approx O(3)$), for biofilm constituted of (a) homogenous bacterial population, (b) heterogenous bacterial population with less adherent cells at the bottom and (c) heterogenous bacterial population with less adherent cells at the cap. Plots named with superscript $^+$show the variation of normalized cell motility with the biofilm structures, the xy plots named with superscript "show the cumulative cell detachment rate and the second xy plots named with superscript" are biofilm roughness plots for cases (a), (b) and (c). The color bar at the bottom indicates the normalized cell motility for a$^+$, b$^+$ and c$^+$. Ten replicates were carried out for each of the parameter settings.

Table 1
Parameter values used in the biofilm simulations. (*mcs* indicates a Monte Carlo Step, J_{GGH} and N_{GGH} indicate the energy and force units in the GGH domain respectively.).

Parameter	Value
Length of simulation domain, x'	1000 lattice sites
Height of simulation domain, y'	1000 lattice sites
Grid size, Δx	1×10^{-6} m
Frequency interval for fluid flow simulations, Δt	240 s
Real time equivalent of 1 mcs	0.4 s
Total simulation time, t	9000 mcs
Volume of bacteria (Fagerlind et al., 2012), V_{cell}	27×10^{-18} m^3
Volume potential, λ	5 J_{GGH}/m^3
Contact energy between two bacteria of specie 'B', C_{BB}	5–15 J_{GGH}
Contact energy between two bacteria of specie 'B*', $C_{B^*B^*}$	15 J_{GGH}
Contact energy between bacteria of specie 'B' and specie 'B*', C_{BB^*}	15 J_{GGH}
Contact energy between any bacteria specie 'B' and substratum 'S', C_{BS}	3 J_{GGH}
Membrane fluctuation temperature, T_m	10–30 J_{GGH}
Pseudo GGH force, $\overrightarrow{F_{GGH}}$	0–10^5 N_{GGH}
Reynold's Number, *Re*	0.07

counterpart (Fig. 5c). Hence, species B* can destabilise the entire system based on its localization in the biofilm structure. Among all the three cases are compared, the case with B* localization to the cap shows accelerated cell erosion as evident in the detachment plots (Fig. 5a′, b′ and c′). In addition, the calculated roughness of the biofilms (Lewandowski and Beyenal, 2013) is higher for these structures (Fig. 5c″). This observation suggests that accelerated erosion should be due to the lowered adhesion potential between the cells at the top, which in turn varies the biomass thickness leading to increased roughness. Such species (B*) should be localized to the interior core of the biofilm structure, where they can protect themselves from erosion and avoid any unintended structural weakening. In nature, morphologies with strong adherent (B) bacterial species covering the peripheral structure and internalized weak adherent species (B*) should evolve spontaneously in flow systems.

4. Conclusions

A GGH method-based model has been developed to understand the effects of shear stress on bacterial biofilm structures. Although we have assumed a hypothetical fluid in our simulations, based on the flow similarity principle, the velocity and pressure should have the same magnitude for any fluid flowing in the same channel with identical Reynolds number. However, the magnitude of stresses calculated will change for different fluids, depending on the density of the fluid used (lighter or heavier). Therefore, the results from the simulations will be the same for any fluid, provided the magnitude '*O*' of force, $\overrightarrow{F_{GGH}}$, used in the simulations are same. This value of $\overrightarrow{F_{GGH}}$ can be adjusted by changing the k_{GGH} in Eq. (4b). The model is also capable of predicting the formation of streamers under low stress conditions. Streamers are found to form in biofilms experiencing low external force or in biofilms with slow moving cells. Multi-event sloughing has been observed in the simulations, suggesting the presence of multiple inflection points in biofilm strength, based on local cell densities. In addition, cells with different adhesivities were introduced to examine the strength of multi-species heterogenous biofilm systems. From the simulation results, it can be hypothesised that less adherent species should always occur enclosed within the biofilm structure for prolonged survival. In future, this model can be combined with growth dynamics to predict the exact time scale of the detachment events. The model can be further improved to accommodate various other external forces such as scrubbing, impact or attrition to evaluate the efficiency of such methods in industrial biofouling clean-up activities.

Appendix

Fig. A1.

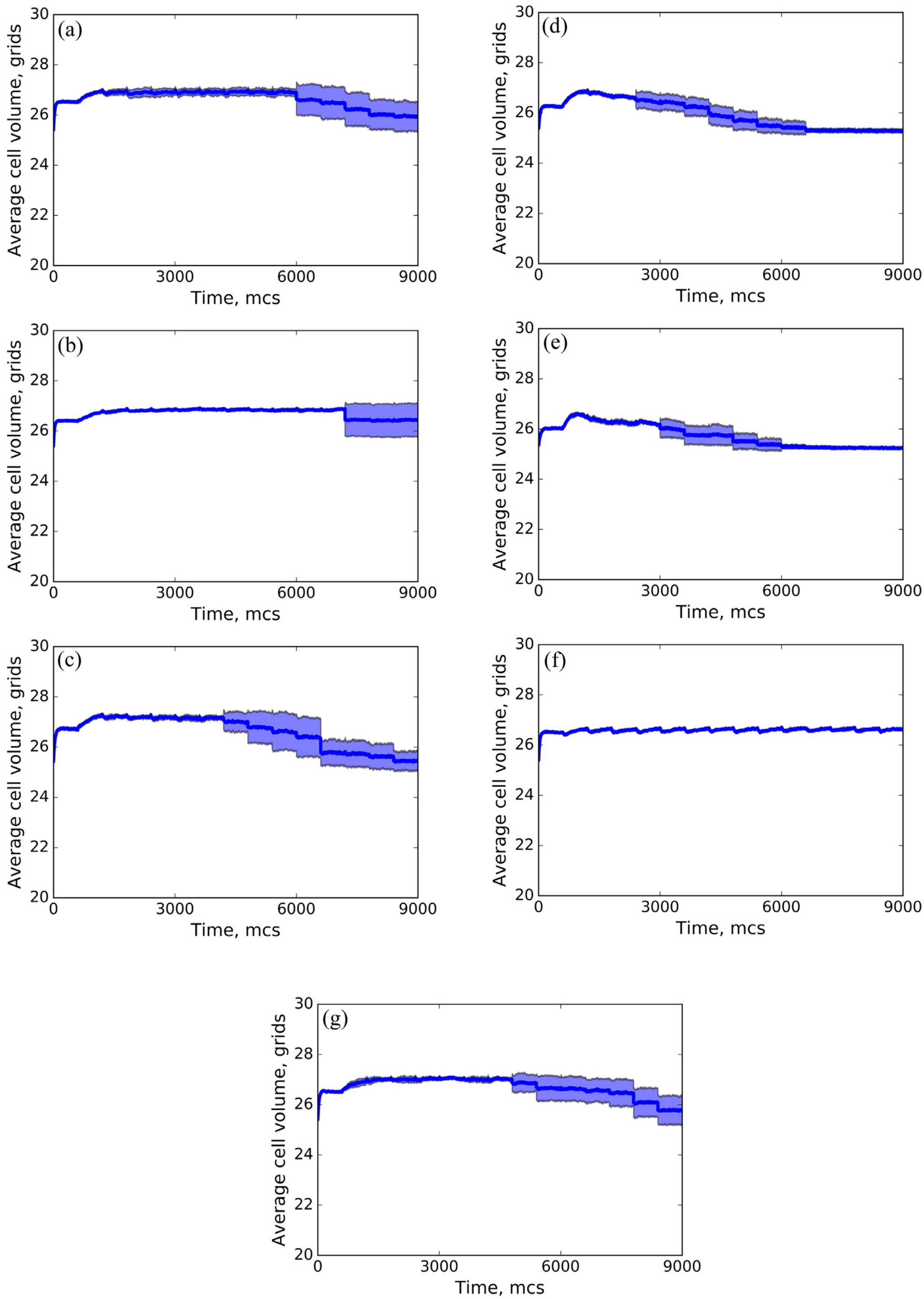

Fig. A1. Variations in average volume of the cells in the simulation for different parametric values. Change in the average volume of cells with $T_m = 20$, C_{BB} =5 and (a, a') $\overrightarrow{F_{GGH}} \approx O(4)$, (b, b') $\overrightarrow{F_{GGH}} \approx O(2)$; $T_m = 20$, $\overrightarrow{F_{GGH}} \approx O(3)$ and (c, c') C_{BB} =10, (d, d') C_{BB} =15; $\overrightarrow{F_{GGH}} \approx O(3)$, C_{BB} =5 and (e, e') $T_m = 10$, (f, f') $T_m = 20$ and (g, g') $T_m = 30$. (The values $T_m = 20$, C_{BB} =5 and $\overrightarrow{F_{GGH}} \approx O(3)$ are used as base sets and one of these values is changed in simulations. Ten replicates were carried out for each of the parameter settings.).

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